Part A :- Conceptual Understanding (Theory)

1. What are Supervised Learning Algorithms?

Supervised Learning Algorithms are machine learning algorithms that learn from labelled data. They use input data and corresponding output values to identify patterns and make predictions. These algorithms are commonly used for tasks such as prediction and classification.

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2. Difference Between Regression Algorithms and Classification Algorithms

Regression algorithms are used to predict continuous numerical values such as house prices, sales, or temperature. Classification algorithms are used to predict categories or classes such as pass/fail, spam/not spam, or yes/no outcomes. The main difference is that regression predicts numbers, while classification predicts labels.

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3. Explain Simple Linear Regression

Simple Linear Regression is a supervised learning algorithm used to find the relationship between one independent variable and one dependent variable. It fits a straight line to the data and uses it to make predictions. The line is represented by the equation:

y=mx+cy = mx + cy=mx+c

where *m* is the slope and *c* is the intercept.

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4. List and Explain the Assumptions of Linear Regression

The main assumptions of Linear Regression are:

- **Linearity:** There should be a linear relationship between input and output variables.

- **Independence:** Observations should be independent of each other.
- **Homoscedasticity:** The variance of errors should remain constant.
- **Normality:** Residual errors should be approximately normally distributed.
- **No Multicollinearity:** Independent variables should not be highly correlated with each other.

These assumptions help improve the reliability and accuracy of the model.

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5. What is Bias–Variance Trade-Off?

Bias–Variance Trade-Off is the balance between underfitting and overfitting in a machine learning model. A model with high bias is too simple and may miss important patterns, while a model with high variance is too complex and may memorize the training data. The goal is to find a balance that provides good prediction performance on new data.

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6. Explain Overfitting and Under fitting with Examples

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary details. As a result, it performs well on training data but poorly on new data. For example, a very complex Polynomial Regression model may over fit house price data.

Under fitting occurs when a model is too simple to capture the relationship in the data. It performs poorly on both training and testing data. For example, using a simple linear model for highly complex house price patterns may lead to under fitting.

Conclusion

Supervised learning algorithms are widely used for prediction and classification tasks. Understanding concepts such as Linear Regression, Bias–Variance Trade-Off, Overfitting, and Under fitting is important for building accurate and reliable machine learning models. These concepts help in selecting the most suitable model and improving prediction performance.